

Model	Final Acc.	Step Acc.	Persona Cons.	Format	Belief-Flip
SFT (Baseline)	73.9	91.5	79.5	98.1	8.3
Vanilla DPO	76.4	92.9	80.1	98.6	10.0
Step-DPO	72.2	90.9	79.1	98.9	10.0
Full-Step-DPO (PRM)	—	—	—	—	—
BC-StepDPO (Type-1 only)	72.2	89.9	81.4	98.1	11.7
Full BC-StepDPO (Type-1 + Type-2)	72.8	91.6	81.7	98.9	10.0
Full BC-StepDPO (Type-2 aug.)	68.9	90.4	83.1	98.1	6.7

Table 1. Ablation across belief-conditional Step-DPO variants (real evaluation). Best per column in bold; proposed methods shaded.